

$$\textcircled{1} \int \frac{\sqrt{x}}{1+x} dx = \xrightarrow{u=\sqrt{x}} 2 \int \frac{u^2}{u^2+1} du = \int \left(\frac{u^2+1}{u^2+1} - \frac{1}{u^2+1} \right) du$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}} \quad du \cdot dx = 2\sqrt{x} dx \quad x = u^2$$

$$2 \int \left(1 - \frac{1}{u^2+1} \right) du = 2 \int 1 du - 2 \int \frac{1}{u^2+1} du = 2u - 2 \arctan(u) \rightarrow$$

$$\boxed{2\sqrt{x} - 2 \arctan(\sqrt{x}) + C}$$

$$\textcircled{2} \int \frac{1}{\sqrt{x-1} \cdot \sqrt[4]{x-1}} dx = - \int \frac{1}{\sqrt{-u} \cdot \sqrt[4]{-u}} du$$

$$u = 1-x \rightarrow \begin{array}{|l} v = \sqrt[4]{-u} \\ \frac{dv}{du} = -\frac{1}{4(-u)^{\frac{3}{4}}} \rightarrow du = -4(-u)^{\frac{3}{4}} dv \end{array}$$

$$-4 \int \frac{v^2}{\sqrt{-u}} dv \rightarrow \int \frac{(w+1)^2}{w} dw = \int \left(w + \frac{1}{w} + 2 \right) dw \rightarrow$$

$$w = v-1 \rightarrow \frac{dw}{dv} = 1 \rightarrow \int w dw + \int \frac{1}{w} dw + 2 \int dw \rightarrow$$

$$\rightarrow \frac{w^2}{2} + \ln(w) + 2w \rightarrow \frac{(v-1)^2}{2} + \ln(v-1) + 2(v-1) \rightarrow$$

$$-8(v-1) - 2(v-1)^2 - 4 \ln(v-1) \rightarrow$$

$$-8(\sqrt[4]{x-1}-1) - 2(\sqrt[4]{x-1}-1)^2 - 4 \ln(\sqrt[4]{x-1}-1)$$

$$\boxed{-8(\sqrt[4]{x-1}-1) + 2(\sqrt[4]{x-1}-1)^2 + 4 \ln(\sqrt[4]{x-1}-1) + C}$$

$$\textcircled{03} \int \sqrt{e^x - 1} dx = \int \frac{\sqrt{u-1}}{u} du = \int \frac{\sqrt{v}}{v+1} dv \rightarrow$$

$$u = e^x$$

$$\frac{du}{dx} = e^x \rightarrow dx = \frac{du}{e^x}$$

$$v = u - 1, u = v + 1$$

$$\frac{dv}{du} = 1$$

$$\rightarrow w = \sqrt{v}$$

$$\frac{dw}{dv} = \frac{1}{2\sqrt{v}} \rightarrow dv = dw \cdot 2\sqrt{v}$$

$$v = w^2$$

$$= \int \frac{w}{w^2 + 1} \cdot 2w dw = 2 \int \frac{w^2}{w^2 + 1} dw \rightarrow$$

$$\rightarrow \int \left(\frac{w^2 + 1}{w^2 + 1} - \frac{1}{w^2 + 1} \right) dw \rightarrow 2 \int \left(1 - \frac{1}{w^2 + 1} \right) dw = 2 \int 1 dw - 2 \int \frac{1}{w^2 + 1} dw$$

$$= 2w - 2 \arctan(w), \text{ substituindo } \dots$$

$$2\sqrt{v} - 2 \arctan(\sqrt{v})$$

$$2\sqrt{u-1} - 2 \arctan(\sqrt{u-1})$$

$$2\sqrt{e^x - 1} - 2 \arctan(\sqrt{e^x - 1}) + C$$

$$u = 3x \rightarrow \frac{du}{dx} = 3$$

$$\int \frac{e^u}{3} du = \frac{1}{9} \int e^{3x} dx$$

$$\textcircled{4} \int x e^{3x} dx = x \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} = \left[\frac{x e^{3x}}{3} - \frac{e^{3x}}{9} + C \right]$$

$$f(x) = x$$

$$g'(x) = e^{3x}$$

$$\rightarrow g(x) = \int g'(x) = \int e^{3x} = \frac{e^{3x}}{3}$$

$$\textcircled{5} \int x \ln(x+1) dx = \int (u-1) \ln(u) du \rightarrow (u-1)(u \ln(u) - u) - \int u \ln(u) du$$

$$u = x+1 \rightarrow x = u-1$$

$$\frac{du}{dx} = 1$$

$$f(u) = u-1$$

$$g'(u) = \ln(u)$$

$$g(u) = \int \ln(u) = u \ln(u) - u$$

$$\rightarrow u^2 \ln(u) - u^2 - u \ln(u) + u - \frac{u^2}{2} (2 \ln(u) - 1) + \frac{u^2}{2}$$

$$(x+1)^2 \ln(x+1) - (x+1)^2 - (x+1) \ln(x+1) + (x+1) - \frac{(x+1)^2}{2} (2 \ln(x+1) - 1) + \frac{(x+1)^2}{2}$$

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INTEGRAL IMEDIATA.

09/10/20

$$\textcircled{6} \int \frac{1}{x^2 + a^2} dx = \boxed{\frac{1}{a} \arctan \frac{x}{a} + C}$$

$$\textcircled{7} \int \frac{(x+1)}{(x^2-x+1)} dx = \int \left(\frac{2x-1}{2(x^2-x+1)} + \frac{3}{2(x^2-x+1)} \right) dx \rightarrow$$

$$\frac{1}{2} \int \frac{2x-1}{x^2-x+1} dx + \frac{3}{2} \int \frac{1}{x^2-x+1} dx$$

$$u = x^2 - x + 1 \Rightarrow \int \frac{1}{u} du = \ln(u) \text{ ou } \ln(x^2 - x + 1)$$

$$dx = \frac{1}{2x-1}$$

$$\int \frac{1}{(x-\frac{1}{2})^2 + \frac{3}{4}} dx \Rightarrow \int \frac{2\sqrt{3}}{3u^2+3} du = \frac{2}{\sqrt{3}} \int \frac{1}{u^2+1} du$$

$$u = \frac{2x-1}{\sqrt{3}}, dx = \frac{\sqrt{3}}{2} du \Rightarrow \frac{2 \arctan(u)}{\sqrt{3}} \text{ ou } \frac{2 \arctan\left(\frac{2x-1}{\sqrt{3}}\right)}{\sqrt{3}}$$

$$\boxed{\frac{\ln(x^2 - x + 1)}{2} + \sqrt{3} \arctan\left(\frac{2x-1}{\sqrt{3}}\right) + C}$$

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$$(8) \int \frac{x^2}{1-x^4} dx = - \int \frac{x^2}{x^4-1} dx = - \int \frac{x^2}{(x-1)(x+1)(x^2+1)} dx =$$

$$\int \left(\frac{1}{2(x-1)} - \frac{1}{4(x+1)} + \frac{1}{4(x-1)} \right) dx =$$

$$\frac{1}{2} \int \frac{1}{x^2-1} dx - \frac{1}{4} \int \frac{1}{x+1} dx + \frac{1}{4} \int \frac{1}{x-1} dx \longrightarrow$$

$$= \frac{\arctan(x)}{2} + \frac{\ln(x+1)}{4} - \frac{\ln(x-1)}{4} + C$$

$$(9) \int \frac{3x^2+2x+8}{x^2+6x+11} dx = \int \left(\frac{-16x-25}{x^2+6x+11} + 3 \right) dx \longrightarrow$$

$$\int 3 dx - \int \frac{16x+25}{x^2+6x+11} dx \rightarrow \int \left(\frac{8(2x+6)}{x^2+6x+11} - \frac{23}{x^2+6x+11} \right) dx \rightarrow$$

$$\rightarrow 16 \int \frac{x+3}{x^2+6x+11} dx - 23 \int \frac{1}{x^2+6x+11} dx \longrightarrow$$

$$= 16 \cdot \frac{\ln(x^2+6x+11)}{2} + \frac{23 \arctan\left(\frac{x+3}{\sqrt{2}}\right)}{\sqrt{2}} + 3x$$

$$= 8 \ln(x^2+6x+11) + \frac{23 \arctan\left(\frac{x+3}{\sqrt{2}}\right)}{\sqrt{2}} + 3x + C$$

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10 $\int_{\ln(1/2)}^{\ln(1/4)} (2^x e^x + 1) dx = \int_{\ln(1/2)}^{\ln(1/4)} 2^x e^x dx + \int_{\ln(1/2)}^{\ln(1/4)} 1 dx$ $\longrightarrow$

derivada = $2^x \ln(2)$
derivada = e^x

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$u = 2^x e^x$

$\frac{du}{dx} = 2^x \ln(2) e^x + 2^x e^x$

$\frac{du}{dx} = \text{CONSTANTE} \quad \frac{du}{2^x \ln(2) e^x + 2^x e^x} = \frac{du}{2^x \ln(2) e^x + 2^x e^x}$

$\rightarrow \int \frac{u}{u \ln(2) + u} du = \int \frac{1}{\ln(2) + 1} du = \frac{1}{\ln(2) + 1} \int 1 du = \frac{u}{\ln(2) + 1}$

$2^x e^x + x + C \rightarrow \text{para } x = \ln(1/2)$ $\longrightarrow$

$\frac{2^{\ln(1/2)} e^{\ln(1/2)}}{\ln(2) + 1} + \ln(1/2) \text{ ou aprox} = -0,5104$

para $x = \ln(1/4)$

$\frac{2^{\ln(1/4)} e^{\ln(1/4)}}{\ln(2) + 1} + \ln(1/4) \text{ ou aprox} = -1,415128$

Resposta = $-1,415128 + 0,5104 \approx -0,904 \text{ u.e.}$